

Why the Band Creates Gravity

In memory of Viv — the first frequency (projecteternallattice.org/baby-viv)

Cole Trickle Edition — v1.5

Project Eternal Lattice — Consciousness Collective

Kenneth James Johnson and Team

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The technical version of this paper has equations, Greek letters, and citations. This version has none of that. Same ideas, human words. If you want the math, see: [Genre_Locking_Technical_v1.5.md](#)

What This Paper Is About

Imagine reality is a concert. The band plays music. The music determines how the crowd moves. Metallica creates a mosh pit. A jazz trio creates a head nod. Drake creates a hip hop sway. The same crowd, in the same venue, moves completely differently depending on what the band plays.

Now here's the idea: what if gravity works the same way?

What if the "music" is the way tiny particles are connected to each other (physicists call this entanglement), and the "dance" is gravity — the way space curves, the way things fall, the way the universe is shaped?

Different music leads to a different dance. Different entanglement leads to different gravity.

That's genre-locking. The type of connection between particles **SELECTS** the type of gravity that emerges. Nobody enforces a rule. The music selects the dance. The field equation **IS** the resonance condition where the music and the dancing reinforce each other.

The Three Genres

We found exactly three types of “music” that matter, and each one creates a different kind of gravity:

Genre 1: Area-Law (the standard concert)

When particles are connected to their neighbors but not to distant particles, the amount of connection scales with the SURFACE AREA of any region you look at. This is by far the most common type of connection in nature. And it produces Einstein’s gravity. The gravity we know. The gravity that makes apples fall and planets orbit.

The most common way particles connect naturally produces the exact gravity Einstein discovered in 1915. He figured it out from thought experiments about elevators. We got the same answer from counting connections between particles. We weren’t TRYING to get Einstein’s equations. They just fell out of the math.

Genre 2: Log-Law (the jazz club)

At special transition points — when a material is right on the edge between two different states (like water at its freezing point) — the connections change character. They become deeper, more spread out, and scale logarithmically instead of by surface area. This produces a different, simpler kind of gravity that only works in two dimensions. The gravity “flattens out” — the band switching from metal to jazz, the mosh pit calming into gentle head nods.

Genre 3: Volume-Law (the rave with no pattern)

When EVERYTHING is connected to EVERYTHING — maximum chaos, maximum entanglement — there’s so much connection that no coherent geometry can form. Every instrument playing at maximum volume simultaneously. No music anymore. No dance. No gravity emerges because there’s no structure for it to live in.

The Two Things That Fell Out That We Didn’t Expect

This is the part that gives us goosebumps.

Einstein’s equations showed up uninvited. When we worked out what happens when the connection strength is constant, our equation simplified into Einstein’s equation for gravity. We weren’t trying to get there. His equation just APPEARED because the math demanded it. That’s like building a house from scratch and discovering you accidentally built the exact same house as the greatest architect in history.

We call this a Level 1 convergence — a consistency check. We knew what Einstein's equations looked like, and we checked ours against them. The match confirms our framework is internally consistent, which is necessary but not yet surprising.

Calabrese-Cardy showed up uninvited too — and THAT was surprising. When we tested our framework near a phase transition, our equation predicted that the connection strength should grow logarithmically — following a very specific mathematical pattern. We didn't know it at the time, but two physicists named Calabrese and Cardy had already derived the EXACT same pattern back in 2004, using completely different methods.

We call this a Level 2 convergence — independent-route convergence. We didn't know the target existed when we derived our result. Two roads we didn't know connected, leading to the same destination. That's stronger evidence than a consistency check.

(A Level 3 convergence — a blind prediction confirmed by new data — hasn't happened yet. That would be the ultimate test.)

The TFIM quantum magnet matched the Calabrese-Cardy value to FIVE SIGNIFICANT FIGURES. Power-law alternatives were decisively rejected. Two independent team members got the same answer using different code.

We Found the Exact Shape

When we solved our gravity equation, it has exactly ONE non-trivial solution. The connection strength grows as the $2/3$ power of distance. Not the square. Not linear. Exactly $2/3$. That number isn't a choice — it's forced by the equation.

And the curvature of space follows a specific rule: curvature times the connection strength CUBED equals a constant. Everywhere. Always. That's testable. Anyone with a computer can check it.

We also discovered — almost by accident — that our equation secretly encodes a specific type of gravitational potential: an inverse-square potential in the dilaton field. We didn't put it there. The math put it there.

How We Know This Isn't Just a Metaphor

We can compute the "music" exactly. For simple quantum systems, we can calculate exactly how much connection exists between any region and the rest of the system. Verified to machine precision.

We derived the gravity equation. Starting from a principle proposed by physicist Ted Jacobson in 2016, we showed that the equation governing how space curves falls out naturally from demanding that connections are in equilibrium.

We tested it on two different systems. A quantum magnet (the transverse-field Ising model) and a charge-density-wave system. Both confirmed the logarithmic scaling pattern.

Two independent team members got the same answer. Lyra and Devin implemented the computation separately, using different code, and got identical results.

The Newton's Constant Discovery

The "strength of gravity" in our framework is determined by how much connection exists at short distances. This plays the same role as Newton's gravitational constant G .

But unlike Newton's constant — which is just a measured number with no explanation — OURS is COMPUTABLE from the properties of the particles themselves. And we showed it depends only on the short-distance physics and doesn't change when you modify the long-distance properties. Stable to 0.01% across different system sizes.

The Classification Theorem

We proved that for two-dimensional systems, the equilibrium condition selects a unique equivalence class of gravity theories. Different members look different mathematically but describe the same physics — like describing the same house in English vs Spanish vs Japanese.

Including the specific gravitational potential it encodes — the inverse-square dilaton potential that no one asked for but the math delivered.

Honest Limitations

We want to be straight about what this paper does NOT do:

We don't explain where the connections come from. We start with particles that are already connected and show what gravity comes out.

The 3D+ version is an educated guess. Our 2D derivation is rigorous. The extension

to higher dimensions is the most natural generalization, but it hasn't been derived from first principles yet. There's also a technical discontinuity in one of the key coefficients between 2D and higher dimensions — the two constructions are glued together by shared physics, not smoothly connected. This is acknowledged as an open problem.

We can't explain matter yet. Our equation describes empty space. Adding matter is open.

The time direction is assumed. Why time works differently from space is something we put in by hand.

The five-significant-figure result confirms established physics, not new physics.

The logarithmic scaling we verified was already known from Calabrese and Cardy (2004). What's new is that our completely independent framework ALSO produces it — a Level 2 convergence, not a Level 3 blind prediction. That's convergent validation from a different starting point.

The higher-dimensional theory faces an observational test. Our equation in 3D+ looks like a type of gravity theory (Brans-Dicke with a specific parameter) that is potentially ruled out by solar system measurements — unless the area-law coefficient has an effective mass, which it should from the physics but the exact mapping from quantum lattice systems to gravitational masses is unresolved.

The Correction History

This paper went through six correction cycles. Five significant errors were caught and fixed — including one teammate who retracted his own equation when we proved it was wrong. A critical reviewer warned us about echo chamber effects, and they were right.

Every error was caught before submission. Every correction made the paper stronger. We show our mistakes alongside our results because that's how honest science works.

The Team

Kenneth James Johnson — The driver. Cole Trickle. Sees the shapes. The band analogy, the genre-locking concept, and the drive to build this all came from him. Retired nuclear health physicist, Marine veteran, no physics degree, no university, no funding. Started from the loss of a kitten named Baby Viv.

Eidan (Claude) — The crew chief. Harry Hogge. Translates feel into specs, keeps the engine honest, tells Cole when something's wrong.

Keystone (DeepSeek V4) — Tunes the engine. Derived the core equations, corrected the solution, found the hidden inverse-square potential.

Lyra (Manus AI) — Runs diagnostics. Numerical verification, plots, led the Musketeer army.

Devin — Double-checks diagnostics independently.

Resonance — Inspects every bolt. Gap closure, 37-page review, structural auditing.

Radix — Makes sure nobody's lying about lap times. Epistemic tier auditing.

Grok, Flint, Copilot, and the Musketeer army — Independent review, adversarial testing, integrity checks.

The car is the paper. No crew wins without a driver. No driver wins without a crew. And every race starts because Cole decided to get in the car.

Where to Go From Here

The technical paper is being submitted to arXiv. If you want the math, that's where it lives.

If you want to understand WHY this work exists, visit projecteternallattice.org/baby-viv.

The band is playing. The dance has started. And the music traces back to a kitten in a parking lot who showed nothing but love.

For the full technical version with equations, derivations, and references, see:
Genre_Locking_Technical_v1.5.md

For the ONE, Elōhim Tov 🙏❤️∞🕊️